

LAB 6 TASKS

Task 1:

```
task1.c
~/Downloads

#include <stdio.h>
#include <unistd.h>
#include <string.h>

int main()
{
    int fd[2];
    char write_msg[100];
    char read_msg[100];
    pid_t pid;

    pipe(fd);
    pid = fork();

    if(pid > 0)
    {
        printf("Enter a string: ");
        fgets(write_msg, sizeof(write_msg), stdin);

        close(fd[0]);
        write(fd[1], write_msg, strlen(write_msg) + 1);
        close(fd[1]);
    }
    else
    {
        close(fd[1]);
        read(fd[0], read_msg, sizeof(read_msg));
        close(fd[0]);

        int i;
        for(i = strlen(read_msg) - 1; i >= 0; i--)
        {
            printf("%c ", read_msg[i]);
        }
    }

    return 0;
}
```

```
student@student-OptiPlex-7000:~/Downloads$ gcc task1.c -o task1
student@student-OptiPlex-7000:~/Downloads$ ./task1
Enter a string: hello from parent
```

```
t n e r a p   m o r f   o l l e h student@student-OptiPlex-7000:~/Downloads$
```

Task 2:

- 1) Child will receive the numbers from parent
- 2) If the buffer size is smaller than the input string, then the child will receive truncated message

Task 3:

```
student@student-OptiPlex-7000:~/Downloads$ touch producer.c
student@student-OptiPlex-7000:~/Downloads$ touch consumer.c
student@student-OptiPlex-7000:~/Downloads$
```

```
student@student-OptiPlex-7000:~/Downloads$ gcc producer.c -o prod
student@student-OptiPlex-7000:~/Downloads$ ./prod
Enter two numbers: 5
8
Enter choice (1: Sum, 2: Product): 1
student@student-OptiPlex-7000:~/Downloads$
```

```
student@student-OptiPlex-7000:~/Downloads$ gcc consumer.c -o cons
student@student-OptiPlex-7000:~/Downloads$ ./cons
Result: 13
student@student-OptiPlex-7000:~/Downloads$
```

```
task1.c      x      task2.c      x      producer.c      x      consumer.c      x
1 #include <stdio.h>
2 #include <fcntl.h>
3 #include <sys/stat.h>
4 #include <unistd.h>
5
6 #define FIFO_FILE "/tmp/myfifo"
7
8 int main()
9 {
10     int fd;
11     int num1, num2, choice;
12
13     mkfifo(FIFO_FILE, 0666);
14
15     printf("Enter two numbers: ");
16     scanf("%d %d", &num1, &num2);
17
18     printf("Enter choice (1: Sum, 2: Product): ");
19     scanf("%d", &choice);
20
21     fd = open(FIFO_FILE, O_WRONLY);
22
23     write(fd, &num1, sizeof(int));
24     write(fd, &num2, sizeof(int));
25     write(fd, &choice, sizeof(int));
26
27     close(fd);
28
29     return 0;
30 }
```

```
task1.c      x      task2.c      x      producer.c      x      consumer.c      x
1 #include <stdio.h>
2 #include <fcntl.h>
3 #include <sys/stat.h>
4 #include <unistd.h>
5
6 #define FIFO_FILE "/tmp/myfifo"
7
8 int main()
9 {
10     int fd;
11     int num1, num2, choice, result;
12
13     fd = open(FIFO_FILE, O_RDONLY);
14
15     read(fd, &num1, sizeof(int));
16     read(fd, &num2, sizeof(int));
17     read(fd, &choice, sizeof(int));
18
19     if(choice == 1)
20         result = num1 + num2;
21     else if(choice == 2)
22         result = num1 * num2;
23     else {
24         printf("Invalid choice\n");
25         return 0;
26     }
27
28     printf("Result: %d\n", result);
29
30     close(fd);
31
32     return 0;
33 }
```